

Natural Rubber — STR Production (Country-specific LUC & Peatland)

🕒 Created by	 Laurent
🕒 Created time	February 9, 2026 10:24 PM
☰ Category	Methodology doc
🕒 Last edited by	 Laurent
🕒 Last updated time	April 16, 2026 9:43 PM

Overview

This LCA model covers the production of **STR (Standard Technical Rubber)**, a natural rubber derivative used for outsole manufacturing in the footwear industry. The model captures **country-specific** environmental impacts for 9 country variants plus a GLO market mix, with four key differentiating parameters:

- **Land use change (LUC)** emissions from rubber plantation establishment
- **Peatland drainage** emissions (particularly significant in Southeast Asia)
- **Electricity mix** (country-specific grid carbon intensity)
- **Crop yield** (determines how much plantation area — and associated inputs — is required per kg of rubber)

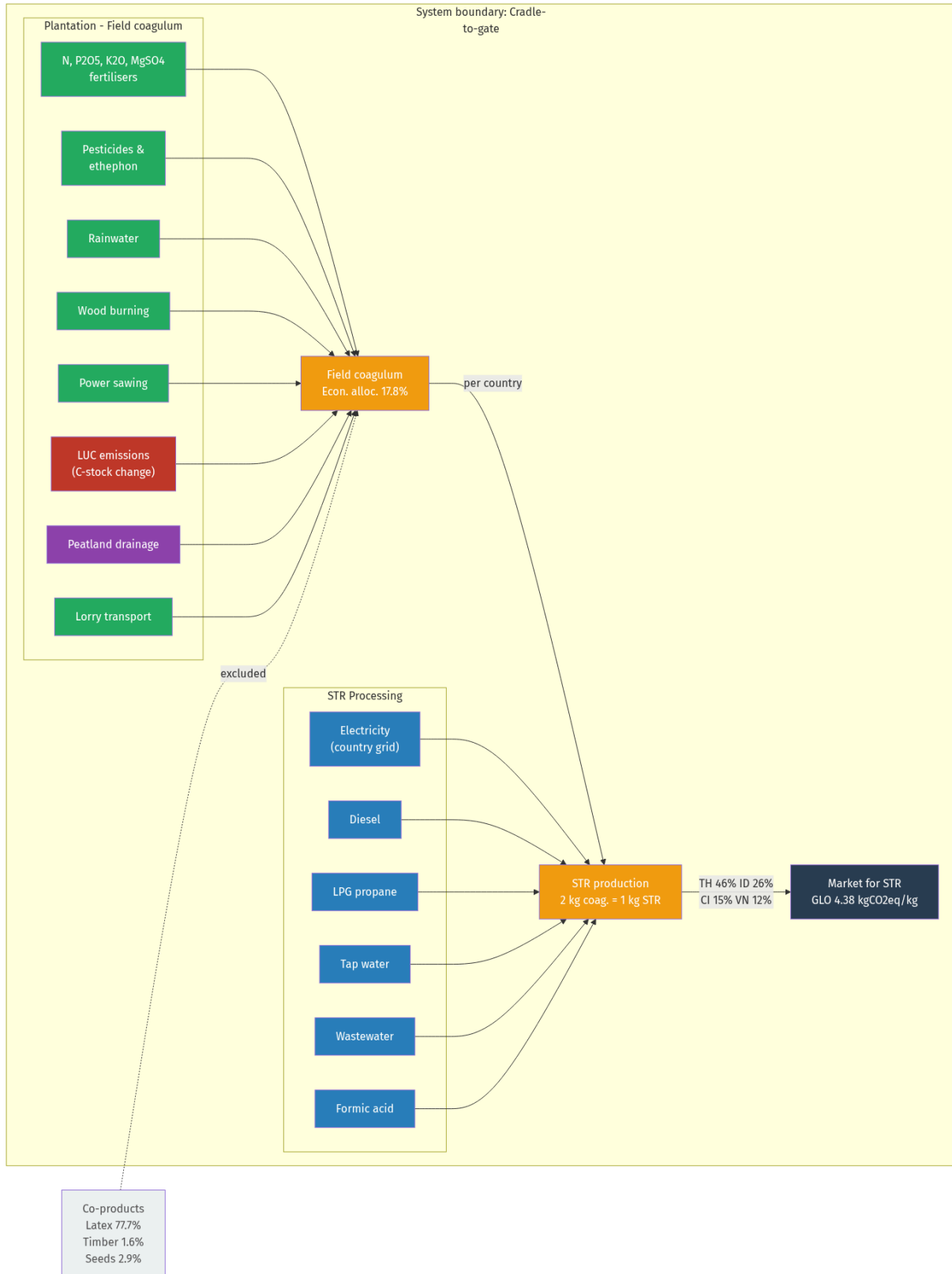
The default GLO market is a **production-weighted mix of the 4 validated countries** (TH, ID, CI, VN). Individual country variants are available for sensitivity analysis.

LCA Design Parameters

- **Functional unit:** 1 kg STR (Standard Technical Rubber)
- **System boundaries:** Cradle-to-gate (plantation through STR processing)
- **Background database:** ecoinvent 3.12 cutoff
- **Impact assessment:** EF v3.1 (16 categories)
- **Brightway project:** Rubber_LUC
- **Foreground database:** rubberdb (from Rubber_study_v2.xlsx)

- **Country variants:** TH, ID, CI, VN + GLO market mix (production-weighted). CN, IN, KH, PH, MY available individually but excluded from GLO (LUC not yet recalculated).

Supply Chain Structure



Color legend: ■ LUC emissions · ■ Peatland drainage · ■ Plantation inputs · ■ STR processing · ■ Core processes · ■ Global market

The model has a 3-level nested activity structure:

1. **Market for STR production (GLO)** — Production-weighted mix of the 4 validated country-specific STR production activities, based on FAOstat 2023 production volumes.
2. **STR production (per country)** — Processing of field coagulum into STR. Based on Jawjit et al. (2010). Inputs: 2 kg field coagulum, 0.22 kWh electricity, 1 MJ diesel, 1.252 MJ LPG (propane), 55 kg tap water, 0.05 m³ wastewater treatment, 0.005 kg formic acid. Country-specific electricity mix.
3. **Field coagulum production (per country)** — Rubber plantation model with economic allocation. Based on Birnbach et al. (2020). Includes individual N, P₂O₅, K₂O, MgSO₄ fertilizers, pesticides, water, wood burning, lorry transport (50 km), and country-specific LUC and peatland drainage emissions.

Plantation model: Field coagulum production

Based on **Birnbach et al. (2020)** — *A condom's footprint: life cycle assessment of natural rubber latex condoms*.

Plantation lifecycle

- **25-year cycle:** planting through felling
- **18.5 years of harvest** (rubber tapping)
- **Economic allocation** applied across co-products

Co-products and economic allocation

Co-product	Price (MYR/kg)	Harvest years	Yearly prod. (kg/ha)	Mass fraction	Economic allocation
Fresh latex	1.84	18.5	2,226	65.3%	77.7%
Field coagulum	2.11	18.5	445	13.1%	17.8%
Timber	0.15	1.0	10,750	17.1%	1.6%
Rubber seeds	1.00	25.0	114	4.5%	2.9%

The **economic allocation factor for field coagulum is 17.8%**, meaning only 17.8% of plantation impacts are attributed to the rubber used for STR production.

Plantation inputs (per 25-year cycle, before allocation)

Input	Total (25 yr)	Unit	Notes
N fertiliser (as N)	897	kg	Inorganic nitrogen fertiliser
P fertiliser (as P ₂ O ₅)	897	kg	Inorganic phosphorus fertiliser
K fertiliser (as K ₂ O)	897	kg	Inorganic potassium fertiliser
Mg fertiliser (MgSO ₄)	97	kg	Magnesium sulfate
Biomass burning (wood waste)	43,500	kg	Plantation clearing/felling
Water	14,000	m ³	Rainwater harvesting
Herbicides	130	kg	Pesticide, unspecified
Fungicides	31	kg	Pesticide, unspecified
Ethephon (growth stimulant)	12.8	kg	Increases latex yield
Power sawing	58.5	machine hours	Timber harvesting
Transport	—	tkm	50 km lorry transport per delivery

STR processing

Based on **Jawjit et al. (2010)** — *Greenhouse gas emissions from rubber industry in Thailand*.

Input	Amount per kg STR	Unit	Notes
Field coagulum	2.0	kg	50% conversion (highly uncertain)
Electricity	0.22	kWh	220 kWh/tonne STR, country-specific grid
Diesel	1.0	MJ	1,000 MJ/tonne STR (agricultural machinery)
LPG (propane)	1.252	MJ	1,252 MJ/tonne STR; modeled as propane at industrial furnace
Tap water	55	kg	Process water
Wastewater treatment	0.05	m ³	Average wastewater
Formic acid	0.005	kg	Coagulation agent

Country-specific Parameters

Key Parameters Overview

Consolidated view of the main differentiating parameters for the 4 countries with fully validated LUC modeling.

Country	Yield (kg/ha/yr)
Thailand	1,440

Country	Yield (kg/ha/yr)
Indonesia	1,188
Ivory Coast	2,000
Vietnam	1,674
CN, IN, KH, PH, MY	⚠ LUC/peatland/yield not yet recalculated. Excluded from GLO market and graphs.
GLO (weighted)	—

GLO market weights (production-based, validated countries only): TH 46.3%, ID 26.0%, CI 15.2%, VN 12.5%.

Yields

Country-specific rubber yields from **FAOstat 2023** determine how much plantation area (and associated inputs) is needed per kg of rubber. Lower yields = more land and inputs per kg = higher impacts.

Data sources

Yield values are from FAOstat 2023 with manual corrections where FAOstat data was unreliable:

- **Thailand** (1,440 kg/ha/yr): Carbonfact sLUC calculator
- **Indonesia** (1,188 kg/ha/yr): FAOstat 2023
- **Ivory Coast** (2,000 kg/ha/yr): Manual override — FAOstat yield was erroneous (~19,706). Derived from APROMAC (2023) and Ecofin Agency (2025) data (1.68 Mt / 722,502 ha, discounted for immature trees)
- **Vietnam** (1,674 kg/ha/yr): FAOstat 2023 (literature alternative: 1,786 from Mekong Resources Governance 2025)

Country	Yield (kg/ha/yr)
Thailand	1,440
Indonesia	1,188
Ivory Coast	2,000
Vietnam	1,674
CN, IN, KH, PH, MY	⚠ Not yet recalculated. Excluded from graphs and GLO market.

The 4 validated countries represent 72% of global NR production.

Land use change & peatland drainage

LUC and peatland drainage emissions are modeled through the **Carbonfact sLUC calculator** — an in-house statistical land use change model that estimates CO₂ emissions from C-stock changes and peatland drainage per country and crop. These are added as direct biosphere CO₂ flows per kg of natural rubber.

LUC methodology

The sLUC (statistical land use change) approach uses FAOStat land area changes (2002–2022) with 20-year linear discounting and proportional attribution of rubber expansion to shrinking land categories. Forest C-stocks use FAO (2020), *Global Forest Resources Assessment 2020*, for AGB/BGB with country-specific SOC sources. Peatland drainage uses a national proxy based on Joosten (2009), *The Global Peatland CO₂ Picture*, and Wetlands International data.

Country	LUC — C-stock change (kgCO ₂ /kg NR)
Thailand	2.02
Indonesia	1.73
Ivory Coast	10.18
Vietnam	4.71
CN, IN, KH, PH, MY	⚠ Not yet recalculated. Excluded from graphs and GLO market.

▼ Thailand — LUC detail

Result: 2.07 kgCO₂/kg NR (C-stock change 2.02 + peatland 0.05). Note: the LUC table above shows C-stock change only (2.02); peatland is an additional 0.05. Yield: 1,440 kg/ha/yr.

The lowest total LUC. Most rubber plantations are 60–80 years old on formerly agricultural land. Only ~15% of recent expansion is attributed to forest. The sLUC result is consistent with an established, mature rubber sector with limited ongoing deforestation.

Comparison with Jawjit (2010) dLUC: 6.4 kgCO₂eq/kg — the 3× gap is driven by:

1. **Land transition scope:** Jawjit assumes 100% forest → rubber; Carbonfact weights actual FAOStat transitions (only 15% from forest, 42% from existing cropland)
2. **Forest C-stocks:** Jawjit 379 tC/ha (IPCC dense rainforest default) vs Carbonfact 220 tC/ha (FAO FRA 2020 + Bridhikitti 2017 MDF SOC)
3. **Rubber C-stocks:** Jawjit 200 tC/ha (Sri Lankan peak maturity) vs Carbonfact 113.8 tC/ha (cross-age Thai averages from Jirakajohnkool 2025, Hytönen 2018)

Carbon stocks used:

Pool	Forest (tC/ha)	Rubber (tC/ha)	Source
AGB+BGB	83.1	66.9	FAO FRA 2020 / Jirakajohnkool 2025
SOC	136.9	46.9	Bridhikitti 2017 MDF
Total	220	113.8	

Net C-stock loss: 106 tC/ha.

Caveats: sLUC is highly uncertain; C-stock values are fragile (no single consolidated source for Thai forest + rubber); most Thai rubber farms are 60–80 yr old with no remaining LUC; Jawjit's 6.4 is a worst-case 100% forest scenario, not an average.

▼ Indonesia — LUC detail

Result: 10.92 kgCO₂/kg NR (C-stock change 1.73 + peatland 9.19). Yield: 1,188 kg/ha/yr.

The highest total, but **84% comes from the peatland drainage proxy**, not from C-stock change. The C-stock change alone (1.73) is comparable to Thailand.

Peatland drainage (9.19 kgCO₂/kg): Calculated as a proxy — total 2008 emissions from peatland drained for agriculture (Joosten 2009 / Wetlands International) divided by total Indonesian agricultural land (~45,817 kha), yielding ~10.9 tCO₂eq/ha/year, divided by the rubber yield (1,188 kg/ha/yr).

Key caveats:

1. **Peatland proxy is rough** — assigns Indonesia's entire agricultural peatland burden equally across all hectares. Rubber on mineral soil (the majority) gets attributed peatland emissions from oil palm-driven drainage.
2. **No peat/mineral soil split** — IPCC 2013 Wetlands Supplement requires different methodologies for organic vs mineral soils. A proper split would dramatically change the result.
3. **Forest SOC was originally missing** — manually corrected, but Indonesian forest SOC on mineral soil (50–70 tC/ha, Guillaume et al. 2018) is lower than Thai MDF SOC (136.9).
4. **Rubber C-stock uses generic "permanent crops" IPCC default** (overestimate at ~102.6 tC/ha) rather than Hevea-specific data.
5. **Peatland source is outdated** (Joosten 2009, pre-2008 only) — doesn't account for post-2008 drainage, peat subsidence, or fire years. Missing CH₄ and N₂O components.

▼ Ivory Coast — LUC detail

Result: 10.21 kgCO₂/kg NR (C-stock change 10.18 + peatland 0.03). Yield: 2,000 kg/ha/yr.

The highest C-stock change of all four countries, driven by:

- **Dense tropical moist forest** with high AGB (150 tC/ha from Lewis et al. 2013) — the highest forest C-stock
- **Direct forest-to-rubber conversion** clearly visible in FAOStat — unlike Vietnam, the sLUC method works well here
- **Rapid recent expansion:** production increased 10× from 164,000 t (2005) to 1.68 Mt (2023), reaching 722,502 ha

Carbon stocks used:

Pool	Forest (tC/ha)	Rubber (tC/ha)	Source
AGB	150.0	40.4 (AGB+BGB)	Lewis et al. (2013), Phil. Trans. R. Soc. B / Blagodatsky 2016
BGB	36.0	(incl. above)	IPCC root-to-shoot 0.24
SOC	47.0	47.0	IPCC 2019, tropical moist Ferralsol / Guillaume 2015
Total	233.0	87.4	

Net C-stock loss: 145.6 tC/ha — the highest of all four countries.

Yield note: FAOStat yield was unreliable (~19,706 kg/ha/yr). Manual yield of 2,000 applied from APROMAC (2023) and Ecofin Agency (2025) data (1.68 Mt / 722,502 ha ≈ 2,322, discounted for immature trees).

Caveats: No CI-specific rubber C-stock data exists (proxies from global meta-analysis); forest AGB is conservative (Lewis reports 336–426 t dm/ha for SW CI); SOC uses IPCC Tier 1 default.

▼ Vietnam — LUC detail

Result: ~5.02 kgCO₂/kg NR (C-stock change 4.71 + peatland 0.31). Yield: 1,674 kg/ha/yr.

Vietnam required a **manual correction** because the standard sLUC method produced an implausibly low result (~0.82 kgCO₂eq/kg) despite having the highest active deforestation risk from rubber. Three structural issues:

1. **"Other land" absorbs ~70% of rubber expansion but has zero C-stock** in IPCC defaults. In reality, Vietnam's "Other land" is recently degraded forest retaining significant biomass and nearly all SOC.
2. **Forest area actually grew** (+379 kha) over 2002–2022 due to reforestation programs and reclassification, masking deforestation in the sLUC model.
3. **Degradation cascade is invisible:** the actual pathway forest → degraded shrubland → rubber appears as two separate events in FAOStat.

Correction applied: "Other land" C-stock set to **80% of forest** (152.1 vs 190.1 tC/ha), justified by:

- Warren-Thomas et al. (2018, Nature Communications): open/degraded forest retains ~54% of biomass C-stock; adding SOC retention, 80% of total ecosystem C-stock is a reasonable estimate
- Result (~5.0 kgCO₂eq/kg) aligns with Warren-Thomas's open forest → rubber scenario for Cambodia (~5.3)

Carbon stocks used:

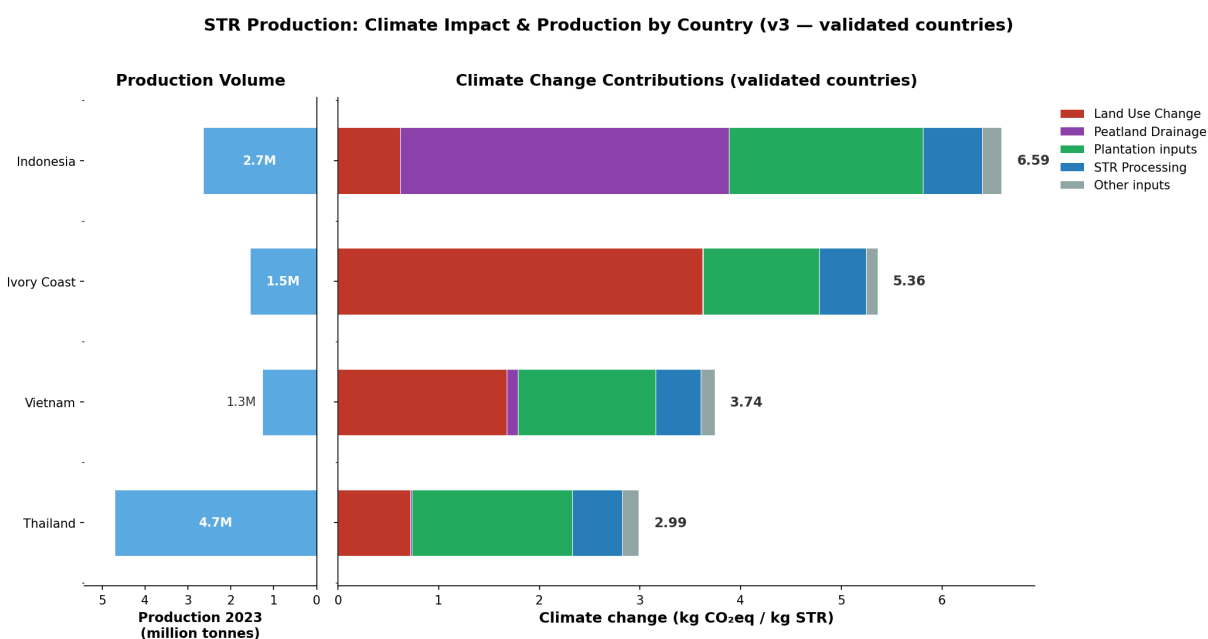
Pool	Forest (tC/ha)	"Other land" (tC/ha)	Rubber (tC/ha)
AGB+BGB	137.1	109.7 (80%)	40.4
SOC	53.0	42.4 (80%)	47.0
Total	190.1	152.1	87.4

Forest: Vietnam FREL/FRL (2016), NFI Cycle 4. SOC: Avitabile et al. (2016). Rubber: Blagodatsky 2016 / Guillaume 2015 (proxies, no Vietnam-specific data).

Caveats: The 80% ratio is expert judgment; sLUC methodology remains structurally unsuited to Vietnam; yield could be updated to 1,786 (Mekong Resources Governance 2025); should be replaced by Approach 3 (geospatial) data when available.

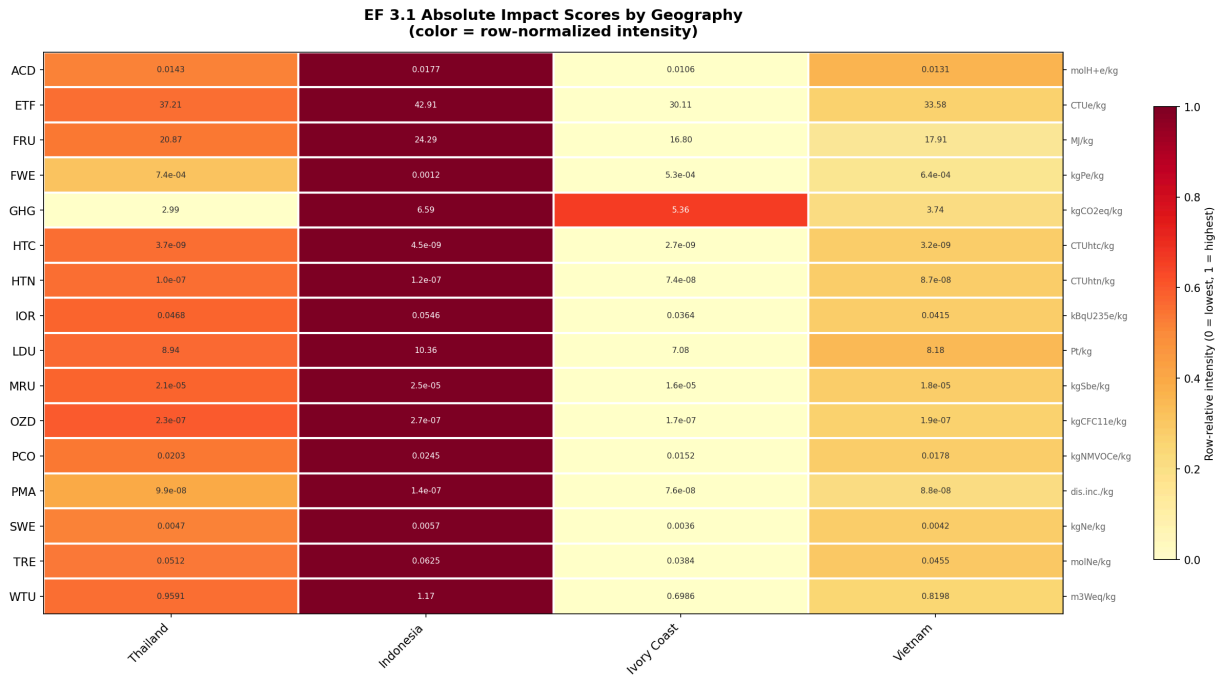
Results Summary

Figure 1: GHG contributions by country (stacked bar + production)



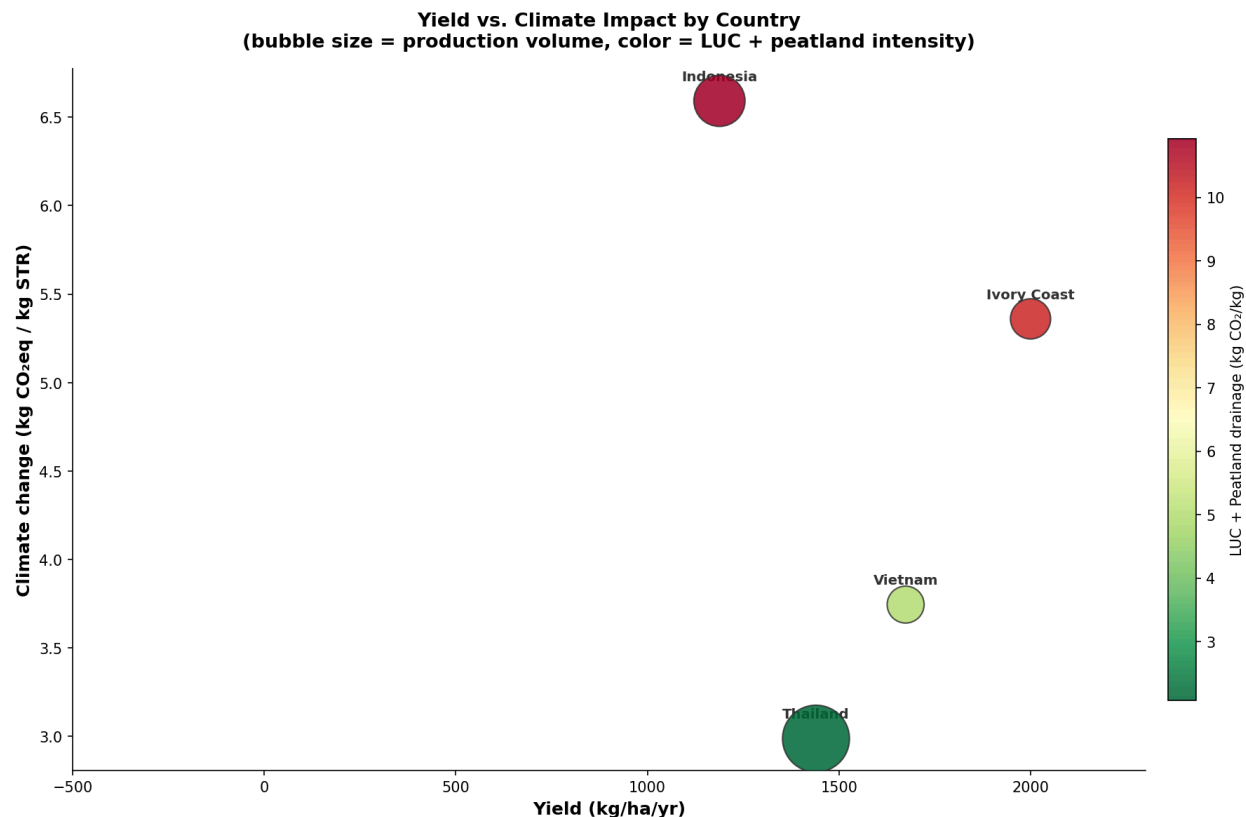
Two-panel figure: left panel shows 2023 production volumes for the 4 validated countries (TH, ID, CI, VN); right panel shows climate change contributions per kg STR, broken down by impact driver (Land Use Change, Peatland Drainage, Plantation inputs, STR Processing).

Figure 2: EF 3.1 heatmap (row-normalized)



Heatmap of all 16 EF 3.1 impact categories across TH, ID, CI, VN. Cell values are absolute scores; color intensity is row-normalized (relative magnitude within each indicator).

Figure 3: Yield vs. climate impact (bubble chart)



Bubble size = production volume, color = LUC + peatland intensity. Shows the relationship between yield and per-kg climate impact.

Key insight: Among the 4 validated countries, Indonesia has the highest per-kg impact (**6.59** kgCO₂eq/kg) driven by peatland drainage (84% of LUC burden), followed by Ivory Coast (**5.36**) from direct deforestation. Vietnam (**3.74**) reflects the corrected degradation cascade. Thailand (**2.99**), the largest producer, now includes LUC from ~15% recent forest conversion. The GLO weighted average is **4.38 kgCO₂eq/kg**.

Key Findings

Indonesia: peatland drainage dominates

Peatland drainage (9.19 kgCO₂/kg) is the dominant contributor to Indonesia's impact, accounting for 84% of the total LUC burden. The C-stock change component (1.73) is comparable to Thailand. Total impact: **6.59 kgCO₂eq/kg STR**.

Ivory Coast: highest deforestation signal

Ivory Coast has the highest C-stock change (10.18 kgCO₂/kg), driven by direct conversion of dense tropical forest to rubber. Total impact: **5.36 kgCO₂eq/kg STR**.

Vietnam: corrected for degradation cascade

Vietnam's LUC (4.71 kgCO₂/kg) required a correction because FAOStat's land categories mask the forest → degraded land → rubber pathway. Total impact: **3.74 kgCO₂eq/kg STR**. See LUC detail toggle for full methodology.

Thailand: largest producer, moderate impact

Despite being the world's largest producer (32.5% of global output), Thailand has a moderate per-kg climate impact (**2.99 kgCO₂eq/kg STR**). LUC of 2.02 kgCO₂/kg comes from ~15% recent forest conversion, mostly in eastern/NE plantations.

Plantation inputs

Individual N, P₂O₅, K₂O, and MgSO₄ fertilizers are modeled separately, revealing that **fertilizer production is a significant share of total GHG** in low-LUC countries (e.g. Thailand).

Electricity mix matters for FWE

Freshwater eutrophication (FWE) is influenced by the electricity mix, with higher impacts in countries where coal has a larger share (e.g. Indonesia).

CN, IN, KH, PH, MY: not yet recalculated

These 5 countries still use preliminary LUC values and yields that have not been recalculated with the Carbonfact sLUC calculator. They are excluded from graphs and the GLO market mix. They remain available for individual queries.

Limitations & Caveats

1. **Single literature source for plantation model** — The entire plantation inventory (inputs, co-product ratios, yields per harvest year) is based on Birnbach et al. (2020), which studied Malaysian rubber plantations. Actual practices vary significantly between countries.
2. **Economic allocation sensitivity** — The 17.8% allocation factor for field coagulum depends on market prices for latex, coagulum, timber, and seeds. Price volatility could shift this allocation significantly.
3. **2 kg coagulum per kg STR** — The 50% conversion factor is from Jawjit et al. (2010) and is noted as “highly uncertain” in the model. This directly scales all upstream plantation impacts.
4. **LUC and peatland values are point estimates** — Country-level averages from the LUC tool mask significant within-country variation (e.g., not all Indonesian rubber is on peatland).
5. **FAOstat yield data** — Some FAOstat yield values required correction due to unit inconsistencies (hg/ha vs kg/ha). Ivory Coast yield was manually overridden from

erroneous FAOStat data.

6. **No uncertainty analysis** — All results are deterministic. The wide range of yields and LUC values across countries suggests high overall uncertainty.
7. **Fertilizer split assumption** — NPK is split into individual N, P₂O₅, K₂O, and MgSO₄ based on nutrient ratios. Actual fertilizer blends and application rates may differ by country and plantation age.
8. **STR processing water and chemicals** — Tap water (55 kg/kg), wastewater (0.05 m³/kg), and formic acid (0.005 kg/kg) amounts are based on industry averages and may vary by facility.

Reference

Scientific Literature

- **Birnbach, M., Lehmann, A., Naranjo, E., & Finkbeiner, M. (2020).** A condom's footprint - life cycle assessment of a natural rubber condom. *The International Journal of Life Cycle Assessment*, 25, 964–979. <https://doi.org/10.1007/s11367-019-01701-y>
- **Jawjit, W., Kroeze, C., & Rattanapan, S. (2010).** Greenhouse gas emissions from rubber industry in Thailand. *Journal of Cleaner Production*, 18(5), 403–411. <https://doi.org/10.1016/j.jclepro.2009.12.003>
- **Nunes, F. M., Silva, A. L. E., May, J., Szarblewski, M. S., Flemming, L., Assmann, E. E., Moraes, J. A. R., & Machado, E. L. (2025).** Environmental impacts associated with the life cycle of natural rubbers: A review and scientometric analysis. *Industrial Crops and Products*, 224, 120350. <https://doi.org/10.1016/j.indcrop.2024.120350>

LUC methodology references

- **Warren-Thomas, E. M., et al. (2018).** Protecting tropical forests from the rapid expansion of rubber using carbon payments. *Nature Communications*. <https://doi.org/10.1038/s41467-018-03287-9>
- **Lewis, S. L., et al. (2013).** Above-ground biomass and structure of 260 African tropical forests. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 368.
- **Blagodatsky, S., & Xu, J. (2016).** Carbon balance of rubber (*Hevea brasiliensis*) plantations: A review of uncertainties at plot, landscape and production level. *Agriculture, Ecosystems & Environment*, 221, 8–19.
- **FAO. (2020).** *Global Forest Resources Assessment 2020: Main report*. Rome: Food and Agriculture Organization of the United Nations.

- **Guillaume, T., et al. (2015).** Losses of soil carbon by converting tropical forest to plantations: erosion and decomposition estimated by $\delta^{13}\text{C}$. *Global Change Biology*.
- **Guillaume, T., et al. (2018).** Carbon costs and benefits of Indonesian rainforest conversion to plantations. *Nature Communications*.
- **Joosten, H. (2009).** *The Global Peatland CO₂ Picture: Peatland status and drainage related emissions in all countries of the world*. Wetlands International, Ede.
- **Jirakajohnkool, S., et al. (2025).** Age-Based Biomass Carbon Estimation and Soil Carbon Assessment in Rubber Plantations Integrating Geospatial Technologies and IPCC Tier 1–2 Guidelines. *Forests*, 16(11), 1652.
- **Hytönen, J., Kaakkurivaara, N., & Kaakkurivaara, T. (2018).** Biomass equations for rubber tree (*Hevea brasiliensis*) components in Southern Thailand. *Journal of Tropical Forest Science*.
- **Bridhikitti, A. (2017).** Soil and biomass carbon stocks in forest and agricultural lands in tropical climates. *Songklanakarin Journal of Science and Technology*, 39(6), 697–707.
- **Government of Viet Nam. (2016).** *Viet Nam's submission on reference levels for REDD+ results-based payments under the UNFCCC*. UNFCCC REDD+ submission.
- **Avitabile, V., Herold, M., Heuvelink, G. B. M., et al. (2016).** An integrated pan-tropical biomass map using multiple reference datasets. *Global Change Biology*, 22(4), 1406–1420. <https://doi.org/10.1111/gcb.13139>

Statistical Data

- **FAO. (2023).** *FAOSTAT: Crops and livestock products*. Food and Agriculture Organization of the United Nations. Accessed 14 April 2026. <https://www.fao.org/faostat/en/#data/QCL>

Industry and sector data

- **APROMAC. (2023).** *Annual report on rubber production in Côte d'Ivoire*. Abidjan, Côte d'Ivoire.
- **Ecofin Agency. (2025, December 24).** Côte d'Ivoire plans major expansion of natural rubber production capacity. <https://www.ecofinagency.com/news/2412-51631-cote-d-ivoire-plans-major-expansion-of-natural-rubber-production-capacity>

Internal Tools

- **Carbonfact sLUC calculator.** Internal statistical land use change model using FAOStat land area transition data (2002–2022) with 20-year linear discounting and crop-specific proportional attribution. Carbon stocks from FAO FRA 2020, IPCC 2019, national FRELs, and

peer-reviewed literature. Peatland drainage from Joosten (2009) / Wetlands International national proxies. Yields from FAOStat with manual QA corrections.